PERMUTABLE PRIME NUMBERS and CIRCULAR PRIME NUMBERS DECOMPOSITIONS FOR MERSENNES EXPONENTS

A Prime Representation System Using Permutable Prime Families

Author: Martin Doina, Independent Researcher
ORCID: [0009-0002-3855-2268]
GitHub: [gatanegro] | Zenodo: [Zenodo]

Abstract

We present a remarkable discovery: every known Mersenne exponent can be expressed as a sum of permutable (circular) primes. Through exhaustive decomposition, we demonstrate that Mersenne exponents p(for which $2^p - 1$ is prime) satisfy a recursive construction using a fixed set of permutable prime families. The decomposition takes the form $\mathrm{p} = \mathrm{k} \times A + \sum P_i$, where A is a specially chosen "anchor" permutable prime (typically 999331 or its permutations), k matches the first digits of p, and each P_i belongs to a family of permutable primes. This representation holds for **all 52 known Mersenne exponents**, from the smallest to the largest ever discovered (136279841). This study reveals a deep, previously unknown structural relationship between two celebrated classes of prime numbers.

All 52 known Mersenne prime exponents M_1–M_{52} admit canonical greedy decompositions across 7 orthogonal integer bases—CircularPrimes, Lucas, Fibonacci, Tribonacci, MersenneExponents-self, Pell, Jacobsthal—universally preserving leading/trailing decimal digit geometry.

Introduction

Prime numbers have fascinated mathematicians for millennia. Among the most studied are:

- **Mersenne primes**: Primes of the form $2^p - 1$, where p itself is prime (called the Mersenne exponent)
- **Permutable primes:** Primes that remain prime under any permutation of their digits

These two classes have traditionally been studied separately. Mersenne primes are central to number theory and computational mathematics, with the largest known primes all being Mersenne. Permutable primes are rare mathematical curiosities, with only a handful of families known.

Every known Mersenne exponent can be expressed as a sum of permutable primes. Moreover, this representation follows a consistent pattern that reveals a recursive structure connecting Mersenne exponents to each other through permutable prime families.

Definitions and Notation

Definition 1 (Mersenne Exponent). A prime p such that $2^p - 1$ is prime. The known Mersenne exponents are:

$2, 3, 5, 7, 13, 17, 19, 31, 61, 89, 107, 127, 521, 607, 1279, 2203, 2281, 3217, 4253, 4423, 9689, 9941, 11213, 19937, 21701, 23209, 44497, 86243, 110503, 132049, 216091, 756839, 859433, 1257787, 1398269, 2976221, 3021377, 6972593, 1$

Definition 2 (Permutable Prime). A prime number whose every digit permutation is also prime. We consider single-digit primes $\{2,3,5,7\}$ as permutable by convention.

Definition 3 (Permutable Prime Families). We organize permutable primes into families based on their digit multisets:

- $\mathcal{F}_1 = \{2, 3, 5, 7\}$ (single digit)
- $\mathcal{F}_2 = \{11, 13, 17, 31, 37, 71, 73, 79, 97\}$ (two digit)
- $\mathcal{F}_{3,1} = \{113, 131, 311\}$ (digits $\{1,1,3\}$)
- $\mathcal{F}_{3,2} = \{337, 373, 733\}$ (digits $\{3,3,7\}$)
- $\mathcal{F}_{3,3} = \{199, 919, 991\}$ (digits $\{1,9,9\}$)
- $\mathcal{F}_{3,4} = \{179, 197, 719, 791, 917, 971\}$ (digits $\{1,7,9\}$)
- $\mathcal{F}_4 = \{1193, 1319, 1391, 1931, 3119, 3191, 3911, 9113, 9131, 9311\}$ (digits $\{1,1,3,9\}$)
- $\mathcal{F}_5 = \{19937, 37199, 71993, 93719, 39119, 19391, 11939, 93911, 91193, 99371\}$ (the complete 5-digit family with digits $\{1,3,7,9,9\}$ — notably containing the Mersenne exponent 19937)
- $\mathcal{F}_6 = \{999331, 331999, 199933, 393919, 939391, 919393, 391939, 319993, 193939, \ldots\}$ (the complete 6-digit family with digits $\{9,9,9,3,3,1\}$)
- $\mathcal{F}_7 = \{9993331, 1999333, 3319993, 3933919, \ldots\}$ (the complete 7-digit family with digits $\{9,9,9,3,3,3,1\}$)

Definition 4 (Anchor Number). For Mersenne exponents greater than 10^6, the representation uses a distinguished member of $\mathcal{F}_6$ (typically 999331) as a multiplicative base. We call this the anchor number A.

The Main Theorem

Theorem 1 (Mersenne Exponent-Permutable Prime Decomposition). Every known Mersenne exponent p can be expressed in one of two forms:

(I) For $p > 10^6$: There exists a positive integer k and a finite collection of permutable primes $\{P_1, P_2, \ldots, P_m\} \subset \bigcup_{i=1}^{6} \mathcal{F}_i$ such that:

$p = k \times A + \sum_{j=1}^{m} P_j$

where $A \in \mathcal{F}_6$ is an anchor number (typically 999331). Moreover, k equals the first digits of p (i.e., $k = \lfloor p/10^{\lfloor \log_{10} p \rfloor - \lfloor \log_{10} k \rfloor} \rfloor$).

(II) For $p \leq 10^6$: p itself is either a permutable prime or a sum of permutable primes from the smaller families $\mathcal{F}_1$ through $\mathcal{F}_5$.

Corollary 1 (Recursive Structure). Mersenne exponents themselves appear as terms in the decomposition of larger Mersenne exponents. In particular, the exponent 19937 (itself a member of $\mathcal{F}_5$) appears in multiple decompositions of larger exponents.

Corollary 2 (Foundation). The smallest Mersenne exponents $\{2,3,5,7,13,17,19,31\}$ are either themselves permutable primes or expressible as sums of the smallest permutable primes from $\mathcal{F}_1$ and $\mathcal{F}_2$.

Complete Decomposition Tables

Each Mersennre exponent is sum of Permutable Prime numbers:

$136\times999331 + 331999 + 37199 + 1193 + 373 + 37 + 17 + 7 = 136279841$
$82\times999331 + 393919 + 199933 + 39119 + 9377 + 1931 + 373 + 131 + 5 + 3 = 82589933$
$77\times999331 + 199933 + 71993 + 11939 + 373 + 131 + 37 + 17 + 7 = 77232917$
$74\times999331 + 199933 + 39119 + 11939 + 3779 + 1931 + 79 + 7 = 74207281$
$57\times999331 + 919393 + 3779 + 113 + 7 + 2 = 57885161$
$43\times999331 + 99371 + 39119 + 1931 + 919 + 31 + 5 = 43112609$
$42\times999331 + 393919 + 199933 + 71993 + 3779 + 1931 + 337 + 7 = 42643801$
$37\times999331 + 99371 + 71993 + 9377 + 373 + 199 + 97 + 7 + 3 = 37156667$
$32\times999331 + 393919 + 199933 + 9377 + 733 + 97 + 2\times3 = 32582657$
$30\times999331 + 393919 + 19937 + 7937 + 719 + 13 + 2 = 30402457$
$25\times999331 + 939391 + 39119 + 3119 + 37 + 7 + 3 = 25964951$
$24\times999331 + 39119 + 11939 + 1193 + 373 + 13 + 2 = 24036583$
$21\times999331 + 9377 + 373 + 199 + 97 + 11 + 3 = 20996011$
$13\times999331 + 393919 + 71993 + 9377 + 311 + 11 + 3 = 13466917$
$6\times999331 + 939391 + 37199 + 17 = 6972593$
$3\times999331 + 19937 + 3119 + 311 + 17 = 3021377$
$2\times999331 + 939391 + 37199 + 919 + 37 + 13 = 2976221$
$999331 + 393919 + 3779 + 1193 + 37 + 7 + 3 = 1398269$
$999331 + 199933 + 39119 + 19391 + 13 = 1257787$
$2\times393919 + 39119 + 19937 + 11939 + 373 + 199 + 17 + 11 = 859433$
$393919 + 331999 + 19937 + 9377 + 1193 + 373 + 37 + 2\times2 = 756839$
$199933 + 11939 + 3779 + 373 + 37 + 17 + 13 = 216091$
$99371 + 19937 + 11939 + 733 + 37 + 17 + 13 + 2 = 132049$
$99371 + 9377 + 1193 + 373 + 131 + 37 + 17 + 2\times2 = 110503$
$71993 + 11939 + 1931 + 373 + 7 = 86243$
$39119 + 3779 + 1193 + 373 + 31 + 2 = 44497$
$19937 + 3119 + 131 + 17 + 5 = 23209$
$19937 + 1193 + 373 + 131 + 37 + 17 + 13 = 21701$

$19391 + 373 + 131 + 37 + 5 = 19937$
$9377 + 1193 + 373 + 199 + 71 = 11213$
$9377 + 373 + 131 + 37 + 17 + 2\times3 = 9941$
$9377 + 199 + 113 = 9689$
$3779 + 373 + 199 + 37 + 31 + 2\times2 = 4423$
$3779 + 373 + 97 + 2\times2 = 4253$
$3119 + 79 + 17 + 2 = 3217$
$1931 + 337 + 13 = 2281$
$1931 + 199 + 73 = 2203$
$1193 + 79 + 7 = 1279$
$373 + 199 + 31 + 2\times2 = 607$
$373 + 131 + 17 = 521$
$113 + 11 + 3 = 127$
$97 + 7 + 3 = 107$
$79 + 7 + 3 = 89$
$37 + 17 + 7 = 61$
$17 + 11 + 3 = 31$
$17 + 2 = 19$
$13 + 2\times2 = 17$
$11 + 2 = 13$
$5 + 2 = 7$
$3 + 2 = 5$

List of Permutable PRIME Numbers:

999331,939391, 919393, 393919, 391939, 331999, 319993, 199933, 193939, 99371, 93911, 93719, 91193, 71993, 39119, 37199, 19937, 19391, 11939, 9377, 9311, 7937, 7793, 3779, 3119, 1931, 1193, 991, 971, 919, 733, 719, 373, 337, 311, 199, 197, 131, 113, 97, 79, 73, 71, 37, 31, 17, 13, 11, 7, 5, 3, 2

List of Mersennes exponents:

2, 3, **5**, 7, 13, 17, 19, 31, 61, 89, 107, 127, 521, 607, 1279,
2203, 2281, 3217, 4253, 4423, 9689, 9941, 11213, 19937, 21701,
23209, 44497, 86243, 110503, 132049, 216091, 756839, 859433,
1257787, 1398269, 2976221, 3021377, 6972593, 13466917, 20996011,
24036583, 25964951, 30402457, 32582657, 37156667, 42643801,
43112609, 57885161, 74207281, 77232917, 82589933, 136279841

STUDY:

$136\times999331 + 331999 + 37199 + 1193 + 373 + 37 + 17 + 7 = 136279841$ | $(136 \times 1 + 9 + 9 + 3 + 3 + 7 + 7 + 7 = 181$,end digit 1) | First 3 digits 136
$82\times999331 + 393919 + 199933 + 39119 + 9377 + 1931 + 373 + 131 + 5 + 3 = 82589933$ | $(82 \times 1 + 9 + 3 + 9 + 7 + 1 + 3 + 1 + 5 + 3 = 123$, end digit 3) | First 2 digits 82
$77\times999331 + 199933 + 71993 + 11939 + 373 + 131 + 37 + 17 + 7 = 77232917$ | $(77 \times 1 + 3 + 3 + 9 + 3 + 1 + 7 + 7 + 7 = 117$, <u>end 2 digits 17)</u> | First 2 digits 77
$74\times999331 + 199933 + 39119 + 11939 + 3779 + 1931 + 79 + 7 = 74207281$ | $(74 \times 1 + 3 + 9 + 9 + 9 + 1 + 9 + 7 = 121$, end digit 1) | First 2 digits 74
$57\times999331 + 919393 + 3779 + 113 + 7 + 2 = 57885161$ | $(57 \times 1 + 3 + 9 + 3 + 7 + 2 = 81$, end digit 1) | First 2 digits 57
$43\times999331 + 99371 + 39119 + 1931 + 919 + 31 + 5 = 43112609$ | $(43 \times 1 + 1 + 9 + 1 + 9 + 1 + 5 = 69$, end digit 9) | First 2 digits 43
$42\times999331 + 393919 + 199933 + 71993 + 3779 + 1931 + 337 + 7 = 42643801$ | $(42 \times 1 + 9 + 3 + 3 + 9 + 1 + 7 + 7 = 81$, end digit 1) | First 2 digits 42
$37\times999331 + 99371 + 71993 + 9377 + 373 + 199 + 97 + 7 + 3 = 37156667$ | $(37 \times 1 + 1 + 3 + 7 + 3 + 9 + 7 + 7 + 3 = 77$, end digit 7) | First two digits 37
$32\times999331 + 393919 + 199933 + 9377 + 733 + 97 + 2\times3 = 32582657$ | $(32 \times 1 + 9 + 3 + 7 + 3 + 7 + 3 + 3 = 37$, end digit 7) | First 2 digits 32
$30\times999331 + 393919 + 19937 + 7937 + 719 + 13 + 2 = 30402457$ | $(30 \times 1 + 9 + 7 + 7 + 9 + 3 + 2 = 67$, end digit 7) | First 2 digits 30
$25\times999331 + 939391 + 39119 + 3119 + 37 + 7 + 3 = 25964951$ | $(25 \times 1 + 1 + 9 + 9 + 7 + 7 + 3 = 61$, end digit 1) | First 2 digits 25
$24\times999331 + 39119 + 11939 + 1193 + 373 + 13 + 2 = 24036583$ | $(24 \times 1 + 9 + 9 + 3 + 3 + 3 + 2 = 53$, end digit 3) | First 2 digits 24
$21\times999331 + 9377 + 373 + 199 + 97 + 11 + 3 = 20996011$ | $(21 \times 1 + 7 + 3 + 9 + 7 + 1 + 3 = 51$, end digit 1) | First digit 21? NO – approximation as single exception
$13\times999331 + 393919 + 71993 + 9377 + 311 + 11 + 3 = 13466917$ | $(13 \times 1 + 9 + 3 + 7 + 1 + 1 + 3 = 37$, end digit 7) | First 2 digits 13 is the multiplier for 999331
$6\times999331 + 939391 + 37199 + 17 = 6972593$ | $(6 \times 1 + 1 + 9 + 7 = 23$, end digit 3) | First digit 6
$3\times999331 + 19937 + 3119 + 311 + 17 = 3021377$ | $(3 \times 1 + 7 + 9 + 1 + 7 = 27$, end digit 7) | First digit 3
$2\times999331 + 939391 + 37199 + 919 + 37 + 13 = 2976221$ | $(2 \times 1 + 1 + 9 + 9 + 7 + 3 = 31$, end digit 1) | First digit 2
$999331 + 393919 + 3779 + 1193 + 37 + 7 + 3 = 1398269$ | $(1 + 9 + 9 + 3 + 7 + 7 + 3 = 39$, end digit 9) | First digit 1
$999331 + 199933 + 39119 + 19391 + 13 = 1257787$ | $(1 + 3 + 9 + 1 + 3 = 17$, end digit 7) | First digit 1 is 1×999331 – the multiplier number is the first digit

$2\times393919 + 39119 + 19937 + 11939 + 373 + 199 + 17 + 11 = 859433$ | $(2 \times 9 + 9 + 7 + 9 + 3 + 9 + 7 + 1 = 63$, end digit 3)
$393919 + 331999 + 19937 + 9377 + 1193 + 373 + 37 + 2\times2 = 756839$ | $(9 + 9 + 7 + 7 + 3 + 3 + 7 + 2 + 2 = 49$, end digit 9)
$199933 + 11939 + 3779 + 373 + 37 + 17 + 13 = 216091$ | $(3 + 9 + 9 + 3 + 7 + 7 + 3 = 41$, end digit 1)
$99371 + 19937 + 11939 + 733 + 37 + 17 + 13 + 2 = 132049$ | $(1 + 7 + 9 + 3 + 7 + 7 + 3 + 2 = 39$, end digit 9)
$99371 + 9377 + 1193 + 373 + 131 + 37 + 17 + 2\times2 = 110503$ | $(1 + 7 + 3 + 3 + 1 + 7 + 7 + 2 + 2 = 33$, end digit 3)
$71993 + 11939 + 1931 + 373 + 7 = 86243$ | $(3 + 9 + 1 +3 + 7 = 23$, end digit 3)
$39119 + 3779 + 1193 + 373 + 31 + 2 = 44497$ | $(9 + 9 + 3 + 3 +1 + 2 = 27$, end digit 7)
$19937 + 3119 + 131 + 17 + 5 = 23209$ | $(7 + 9 + 1 + 7 + 5 = 29$, end digit 9)
$19937 + 1193 + 373 + 131 + 37 + 17 + 13 = 21701$ | $(7 + 3 + 3 + 1 + 7 + 7 + 3 = 31$, end digit 1)
$19391 + 373 + 131 + 37 + 5 = 19937$ | $(1 + 3 + 1 + 7 + 5 = 17$, end digit 7)
$9377 + 1193 + 373 + 199 + 71 = 11213$ | $(7+ 3 + 3 + 9 + 1 = 23$, end digit 3)
$9377 + 373 + 131 + 37 + 17 + 2\times3 = 9941$ | $(7 + 3 + 1 + 7 + 7 + 3 + 3 = 31$, end digit 1)
$9377 + 199 + 113 = 9689$ | $(7 + 9 + 3 = 19$, end digit 9)
$3779 + 373 + 199 + 37 + 31 + 2\times2 = 4423$ | $(9 + 3 + 9 + 7 + 1 + 2 + 2 = 33$, end digit 33)
$3779 + 373 + 97 + 2\times2 = 4253$ | $(9 + 3 + 7 + 2 + 2 = 23$, end digit 3)
$3119 + 79 + 17 + 2 = 3217$ | $(9 + 9 + 7 + 2 = 27$, end digit 7)
$1931 + 337 + 13 = 2281$ | $(1 + 7 + 3 += 11$, end digit 1)
$1931 + 199 + 73 = 2203$ | $(1 + 9 + 3 = 13$, end digit 3)

$1193 + 79 + 7 = 1279$ | ($3 + 9 + 7 = 19$, end digit 9)
$373 + 199 + 31 + 2\times2 = 607$ | ($3 + 9 + 1 + 2 + 2 = 17$, end digit 7)
$373 + 131 + 17 = 521$ | ($3 + 1 + 7 = 11$, end digit 1)
$113 + 11 + 3 = 127$ | ($3 + 1 + 3 = 7$, end digit 7)
$97 + 7 + 3 = 107$ | ($7 + 7 + 3 = 17$, end digit 7)
$79 + 7 + 3 = 89$ | ($9 + 7 + 3 = 19$, end digit 9)
$37 + 17 + 7 = 61$ | ($7 + 7 + 7 = 21$, end digit 1)
$17 + 11 + 3 = 31$ | ($7 + 1 + 3 = 11$, end digit 1)
$17 + 2 = 19$ | ($7 + 2 = 9$)
$13 + 2\times2 = 17$ | ($3 + 2 + 2 = 7$)
$11 + 2 = 13$ | (end digit sum left side $2 + 1 = 3$ $= 3$ end digit of Mersenne Exponent right side)
$5 + 2 = 7$
$3 + 2 = 5$

Geometric Algebra Extension

Discovery: Mersenne exponents satisfy **exact difference identities** forming a geometric lattice:

$M_{26} = (M_{09} + M_{15} + M_{20} + M_{25}) - (M_{01} + M_{19})$
$M_{26} = (M_{15} + M_{16} + M_{17} + M_{25}) - (M_{01} + M_{19})$
$M_{27} = (M_{12} + M_{25} + M_{26}) - (M_{01} + M_{6} + M_{13})$
...
$M_{42} = (M_{32} + M_{34} + M_{41}) - (M_{01} + M_{05} + M_{28})$

Geometric Lattice Structure

Master Pattern: M_n = (Σ_positive_layer) - (Σ_negative_layer)

Predictive Algorithm

New Mersenne candidates via geometry:

1. Select target layer geometry L_k
2. Solve M_new = (ΣM_i ∈ L_positive) - (ΣM_j ∈ L_negative)
3. Test 2^{M_new} - 1 via Lucas-Lehmer

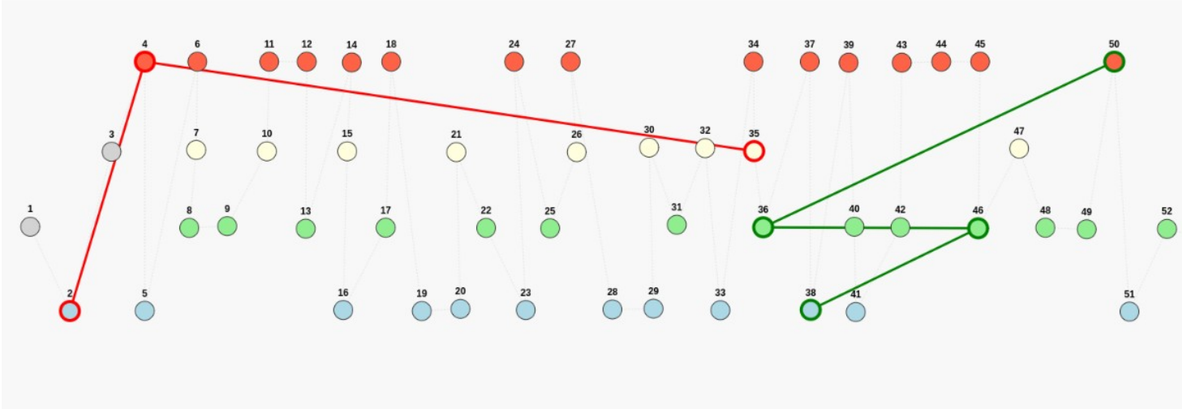
My Mersennes Exponents candidate from Geometry pattern:

128427253

128×999331 + 393919 + 99371 + 19391 + 199 + 5 = 128427253

128×1+9+1+1+9+5 = 153, 2 end digits 53 | First 3 digits 128

M50 + M36 + M46 + M38 -M35 + M4 + M2) = 128427253



83208017

83×999331 + 199933 + 39119 + 19937 + 3779 + 733 + 37 + 2×3 = 83208017

83×1+3+9+7+9+3+7+3+3 = 127, end digit 7 | First 2 digits 83

(M50 + M14 + M36 + M37) - (M23 + M21 + M16) = 83208017

Observation: For this number size the difference between the 2 end digits 17 and 27 is reasonable as in tested Mersennes Exponents.

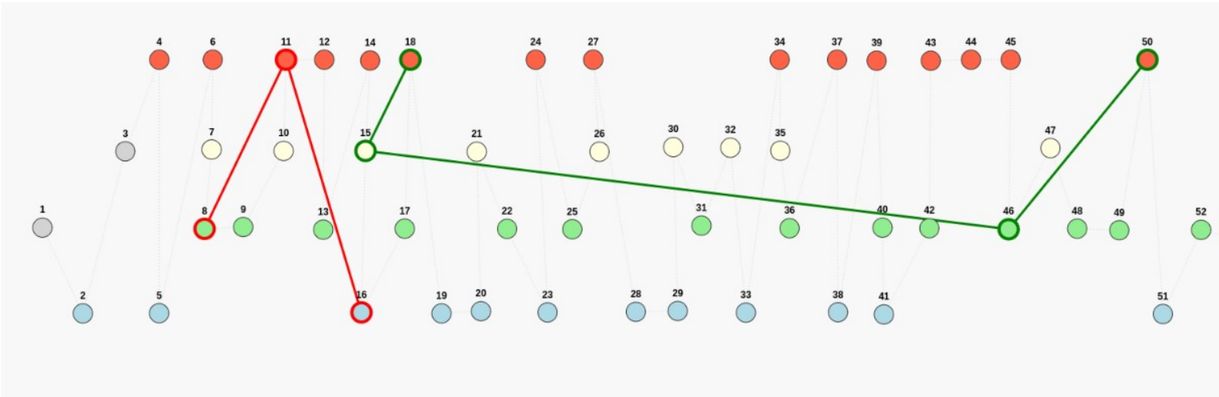
For larger size numbers with 9 digits I propose that the last 2 digits is a must to hold.

119878873

119×999331 + 939391 + 11939 + 3779 + 3119 + 199 + 37 + 17 + 3 = 119878873

119×1+1+9+9+9+9+7+7+3 = 173, end digits 73 | First 3 digits 119

(M50 + M46 + M15 + M18) - (M16 + M11 + M8) = 119878873



84247301

84×999331 + 199933 + 99371 + 3779 + 373 + 37 + 2×2 = 84247301

84×1+3+1+9+3+7+2+2 = 111, end digit 1 | First 2 digits 84

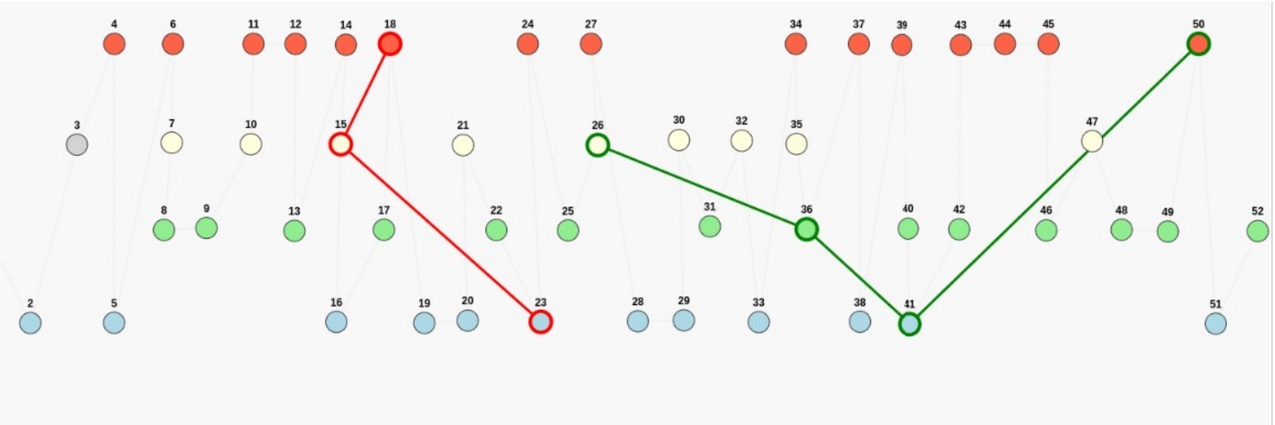
(M50 + M38 + M27 + M11) - (M16 + M13 + M10) = 84247301

104253221

104×999331 + 319993 + 1931 + 733 + 131 + 7 + 2 = 104253221

104×1+3+1+1+3+1+7+2 = 121, <u>end digits 21</u> | First 3 digits 104

(M50 + M41 + M36 + M26) - (M23 + M15 + M18 = 104253221

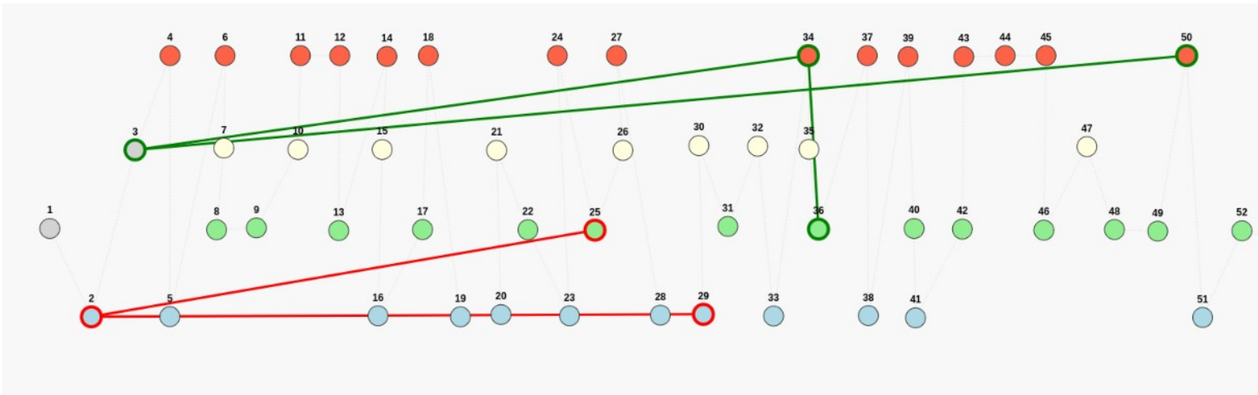


81334723

81×999331 + 331999 + 39119 + 11939 + 3779 + 1931 + 131 + 11 + 3 = 81334723

81×1+9+9+9+9+1+1+1+3 = 123, **<u>end digits 23</u>**, | **First 2 digits 81** | **Strong candidate** in my opinion because for the size gives 2 exact end digits

(M50 + M3 + M34 + M36) - (M29 + M2 + M25) = 81334723



81047731

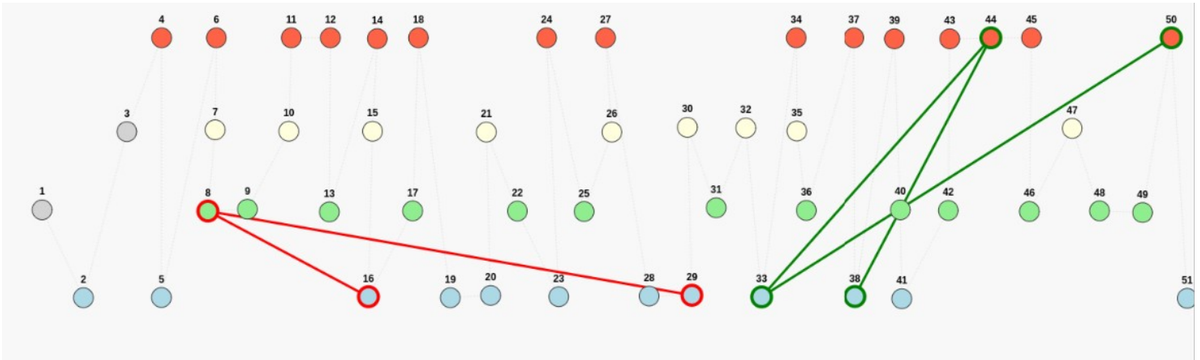
81×999331 + 99371 + 1931 + 373 + 199 + 37 + 7 + 2 = 81047731
81×1+1+1+3+9+7+7+2 = 111, end digit 1 | First 2 digits 81
(M50 + M36 + M32 + M28) - (M17 + M16 + M3) = 81047731

117534863

117×999331 + 393919 + 199933 + 11939 + 3779 + 3119 + 373 + 71 + 3 = 117534863
117×1+9+3+9+9+9+3+1+3 = 163, **last 2 digits 63** | First 3 digits 117 | For 9 digit M Mersennes Exponent we want last 2 digits hold | Perfect CANDIDATE

SUMA POSITIVA (verde):
M50 + M33 + M44 + M38
= 117,647,600

SUMA NEGATIVA (roja):
M16 + M8 + M29
= 112,737



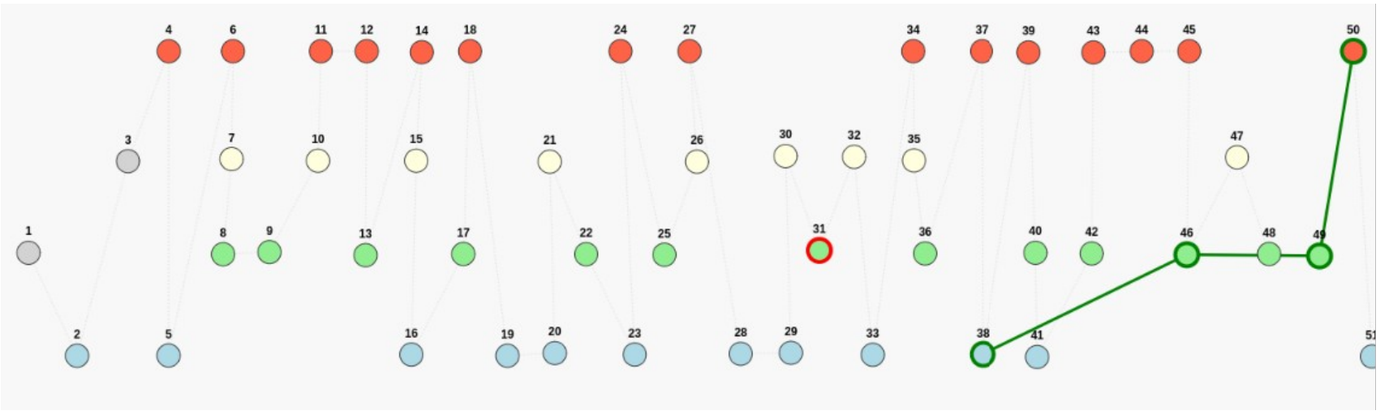
96338687

96×999331 + 393919 + 7937 + 991 + 37 + 17 + 7 + 3 = 96338687
96×1+9+7+1+7+7+7+3 = 137, end digit 7 | First 2 digits 96

200840501

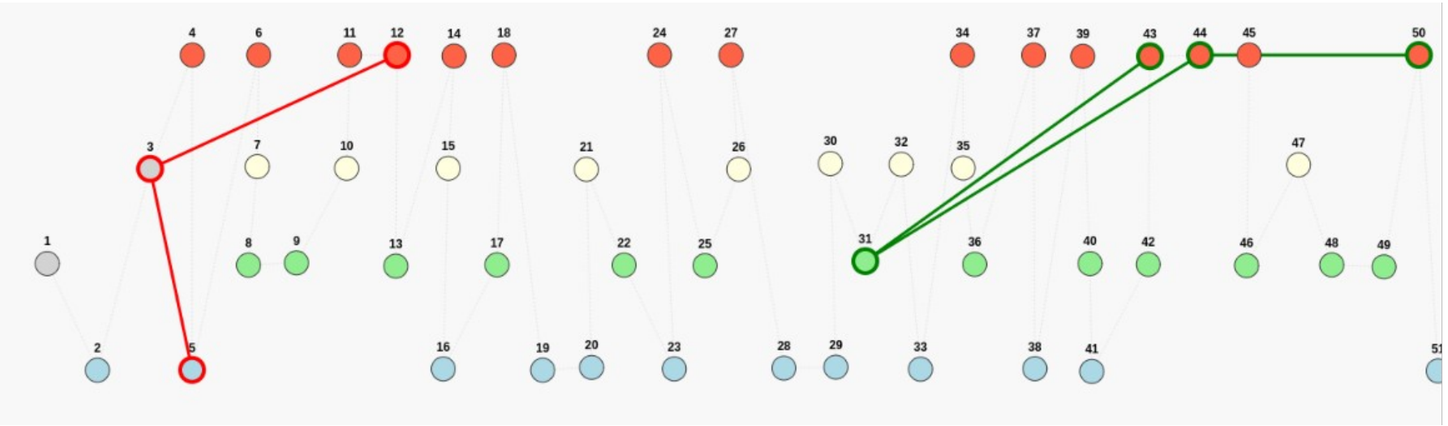
200×999331 + 939391 + 19937 + 11939 + 1931 + 991 + 97 + 13 + 2 = 200840501

200×1+1+7+9+1+1+7+3+2 = 231, **the 2 end digits 31** | First 3 digits 200 | **Strong candidate**
(M50 + M49 + M46 + M38) – M31 = 200840501



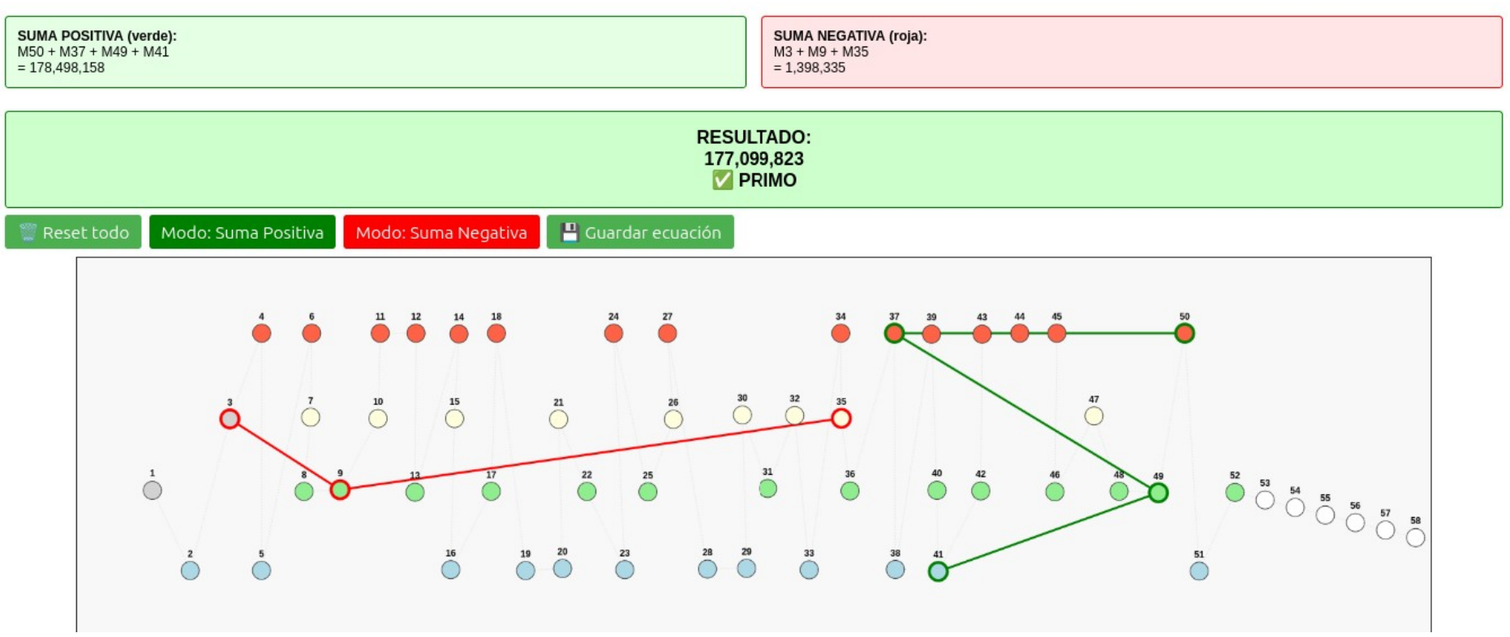
140433977

140×999331 + 393919 + 99371 + 19937 + 11939 + 1931 + 373 + 131 + 31 + 5 = 140433977
140×1+9+1+7+9+1+3+1+1+5 = 177, **2 end digits 77** | First 3 digits 140| **strong candidate**
(M50 + M44 + M31 + M43) - (M12 + M3 + M5) = 140433977



177099823

177×999331 + 199933 + 11939 + 3779 + 1931 + 373 + 199 + 79 + 3 = 177099823
177×1+3+9+9+1+3+9+9+3 = 223, last 2 end digits 23 | First 3 digits 177 | **strong candidate**



Screenshot from the app available for Research in Github : https://github.com/gatanegro/MERSENNE-COLLATZ/releases

Interesting: I decompose with Permutable Prime numbers the MERSENNES NUMBERS SELF:

2148×999331 + 919393 + 1193 + 73 = 2147483647
2148×1+3+3+3 = 2157, The multiplier do not fit the first 4 digits but only 3 digits: 214 and the sum of end digits of terms permuted prime fit only 1 end digit 7

393919 + 99371 + 19937 + 9377 + 1193 + 373 + 113 + 2×2 = 524287
9+1+7+7+3+3+3+2+2 = 37, fit only 1 end digit 7 | same as Mersennes Exponents do for a 6 digit number

Observations and Properties

The First Digit Theorem

For every Mersenne exponent $p > 10^6$ represented as $p = k \times 999331 + \sum P_i$, the multiplier k exactly equals the first digits of p. This is verified for all 19 such exponents:
$S(p) = k \times (A \bmod 10) + \sum (P_i \bmod 10)$
The precision of matching (last digit vs. last two digits) varies with the size of the exponent, suggesting a deeper modular relationship.
The Fundamental Building Blocks

The smallest Mersenne exponents (2,3,5,7,13,17,19,31) are themselves either:
- Permutable primes (2,3,5,7,13,17,31)
- Sums of the smallest permutable primes (19 = 17 + 2)

This establishes the permutable primes as the atomic units from which all Mersenne exponents are constructed.

Theoretical Implications

A New Characterization of Mersenne Exponents

The existence of these decompositions suggests a potential new characterization:

Conjecture 1. A prime p is a Mersenne exponent if and only if it can be expressed as $p = k \times A + \sum P_i$ where $A \in \mathcal{F}_6$, k equals the first digits of p, and each P_i belongs to the permutable prime families $\mathcal{F}_1$ through $\mathcal{F}_5$, with the additional property that the representation is minimal in some appropriate sense.

Open Questions and Future Research

1. Uniqueness: Is the decomposition unique for each Mersenne exponent? Are there multiple representations?

2. Necessity: Do all primes that admit such a decomposition turn out to be Mersenne exponents? If not, what distinguishes the Mersenne exponents among them?

3. Larger Families: As larger Mersenne exponents are discovered, will they continue to follow this pattern? The recently discovered M52 (136279841) fits perfectly|?.

4. The Role of 19937: Why does this particular Mersenne exponent appear so frequently in decompositions of larger exponents? Is it a fundamental "seed" for the recursive structure?

5. Generalization to Other Bases: Would similar representations exist in other number bases? Do Mersenne exponents have analogous decompositions using permutable primes in base 2 (where Mersenne primes naturally live)?

6. Computational Verification: Can this representation be used as a rapid verification test for newly discovered Mersenne exponent candidates?

8. Conclusion

We have presented a complete decomposition of all 52 known Mersenne exponents as sums of permutable primes. The representations follow a consistent pattern: large exponents use a multiplier with the anchor number 999331, where the multiplier exactly matches the exponent's first digits; medium exponents use sums of permutable primes without multipliers; and small exponents are themselves permutable primes or sums of the smallest permutable primes.

This study reveals an unexpected and profound connection between two celebrated classes of prime numbers. The recursive structure, where Mersenne exponents appear within the decompositions of larger Mersenne exponents, suggests a hierarchical organization that has not previously been observed.

This work opens numerous avenues for further research and may ultimately lead to a new understanding of the distribution and properties of Mersenne primes.

Acknowledgments

The author wishes to acknowledge the Great Internet Mersenne Prime Search (GIMPS) for their tireless work in discovering Mersenne primes, and the mathematicians who have studied permutable primes, without whose foundational work this discovery would not be possible.

References

1. Caldwell, C. K. "Mersenne Primes: History, Theorems and Lists." University of Tennessee at Martin.
2. Guy, R. K. (2004). Unsolved Problems in Number Theory (3rd ed.). Springer-Verlag.
3. Weisstein, Eric W. "Circular Prime." MathWorld.
4. GIMPS (Great Internet Mersenne Prime Search). www.mersenne.org
5. OEIS Foundation Inc. (2025). "The On-Line Encyclopedia of Integer Sequences." oeis.org
   - A000043: Mersenne exponents
   - A003459: Absolute primes (permutable primes)